

MATH 3220-002 PSet 7 Specification

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1 Background

This problem set focuses on **multilinear algebra** and polynomial functions. Much like in the single variable setting, polynomial functions in the multivariable setting act as simple models of differentiable functions, in that by way of Taylor approximation any differentiable function is a polynomial function up to higher order terms.

In the multivariable setting, a consistent definition of a polynomial function requires multilinear transformations, which is why this problem set starts with a problem on the algebraic properties of sets of multilinear transformations. The second problem then focuses on the index management for polynomial functions between open subsets of spaces of tuples, introducing the **multi-index notation**. The final two problems engage with polynomial functions as objects within differential calculus.

Two special kinds of multilinear transformations are those that are **symmetric** and **alternating** (defined below). Symmetric multilinear transformations are more relevant for higher order derivatives (and hence for Taylor approximation), whereas we'll encounter and make

use of alternating multilinear transformations later when we start studying **differential forms**.

Symmetric and Alternating Multilinear Transformations Let V and W be vector spaces, k be a nonnegative integer and $T : V^k \rightarrow W$ be a k -linear transformation. T is called **symmetric** if for any $(v_1, v_2, \dots, v_k) \in V^k$ and for any $i, j \in \{1, 2, \dots, k\}$ with $i \neq j$,

$$\begin{aligned} &T(v_1, v_2, \dots, v_{i-1}, v_i, v_{i+1}, \dots, v_{j-1}, v_j, v_{j+1}, \dots, v_k) \\ &= T(v_1, v_2, \dots, v_{i-1}, v_j, v_{i+1}, \dots, v_{j-1}, v_i, v_{j+1}, \dots, v_k). \end{aligned}$$

On the other hand T is called **alternating** if for any $(v_1, v_2, \dots, v_k) \in V^k$ and for any $i, j \in \{1, 2, \dots, k\}$ with $i \neq j$,

$$\begin{aligned} &T(v_1, v_2, \dots, v_{i-1}, v_i, v_{i+1}, \dots, v_{j-1}, v_j, v_{j+1}, \dots, v_k) \\ &= -T(v_1, v_2, \dots, v_{i-1}, v_j, v_{i+1}, \dots, v_{j-1}, v_i, v_{j+1}, \dots, v_k). \end{aligned}$$

We denote by $\mathcal{S}_k(V; W)$ and $\mathcal{A}_k(V; W)$ the collections of all symmetric and alternating k -linear transformations $V^k \rightarrow W$, respectively.

Polynomial Functions Let E and F be normed spaces, $M \subseteq E$ and $N \subseteq F$ be nonempty open subsets, and $p : M \rightarrow N$ be a function. p is said to be a **polynomial function**, if for each $n \in \mathbb{Z}_{\geq 0}$, there is a bounded n -linear transformation $T_n : E^n \rightarrow F$ such that

1. $T_n = 0$ for all but finitely many n , and
2. For any $x \in M$, $p(x) = \sum_{n=0}^{\infty} T_n x^n$, where $T_n x^n = T_n(\underbrace{x, x, \dots, x}_{n \text{ many}})$.

In this case, we call that $T_\bullet$ is a **sequence of bounded multi-linear operators associated to** p . The **degree** d of p is defined to be the largest nonnegative integer n such that $T_n \neq 0$ as n -linear transformations. In this case we write

$$p(x) = \sum_{k=0}^d T_k x^k.$$

2 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details. The **hyperlinks** too are meant to serve as optional additional context; chasing and studying them are by no means necessary to solve the problems. Indeed this problem set is designed so that, broadly speaking, the lectures are sufficient to complete it.

When tackling statements in MATH 3220, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.
2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical **error detection and correction skills**.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example (say as proof to a given existential statement), you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook; it's up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. Let V and W be vector spaces and k be a non-negative integer.

For any two non-negative integers p, q with $p \geq q$, denote the binomial coefficient ("p choose q") by $\binom{p}{q} = \frac{p!}{q!(p-q)!}$.

- (a) The set $\mathcal{L}_k(V; W)$ of all k -linear transformations $V^k \rightarrow W$ is a vector space.
- (b) The set $\mathcal{S}_k(V; W)$ of all symmetric k -linear transformations $V^k \rightarrow W$ is a vector space.
- (c) The set $\mathcal{A}_k(V; W)$ of all alternating k -linear transformations $V^k \rightarrow W$ is a vector space.
- (d) The only k -linear $V^k \rightarrow W$ that is simultaneously symmetric and alternating is the zero transformation.
- (e) $\dim(\mathcal{L}_k(V; W)) = \dim(V)^k \dim(W)$.
- (f) $\dim(\mathcal{A}_k(V; W)) = \begin{cases} \binom{\dim(V)}{k} \dim(W), & \text{if } \dim(V) \geq k \\ 0, & \text{if } \dim(V) < k \end{cases}$.
- (g) $\dim(\mathcal{S}_k(V; W)) = \binom{\dim(V) + k - 1}{k} \dim(W)$.

2. Let $M \subseteq \mathbb{R}^n$, $N \subseteq \mathbb{R}^m$ be nonempty open subsets and $f : M \rightarrow N$ be a function with components $f_1, f_2, \dots, f_m$, so that

$$\begin{aligned} \forall x = (x_1, x_2, \dots, x_n) \in M : f(x) &= (f_1(x), f_2(x), \dots, f_m(x)) \\ &= (f_1(x_1, x_2, \dots, x_n), f_2(x_1, x_2, \dots, x_n), \dots, f_m(x_1, x_2, \dots, x_n)). \end{aligned}$$

Then f is a polynomial function iff there is a function

$$c_\bullet : \{1, 2, \dots, m\} \rightarrow F(\mathbb{Z}_{\geq 0}^n; \mathbb{R})$$

such that for any $j \in \{1, 2, \dots, m\}$,

- $c_j(\alpha_1, \alpha_2, \dots, \alpha_n) = 0$ for all but finitely many $(\alpha_1, \alpha_2, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$,
and
- for any $x = (x_1, x_2, \dots, x_n) \in M$,

$$\begin{aligned}
 f_j(x) &= c_j(0, 0, \dots, 0, 0) \\
 &+ c_j(1, 0, \dots, 0, 0)x_1 + c_j(0, 1, 0, \dots, 0)x_2 + \dots + c_j(0, 0, \dots, 0, 1)x_n \\
 &+ c_j(2, 0, \dots, 0, 0)x_1^2 + c_j(1, 1, 0, \dots, 0)x_1x_2 + \dots + c_j(1, 0, \dots, 0, 1)x_1x_n \\
 &+ c_j(0, 2, 0, \dots, 0)x_2^2 + \dots + c_j(0, \dots, 0, 2)x_n^2 \\
 &+ c_j(3, 0, \dots, 0)x_1^3 + c_j(2, 1, \dots, 0)x_1^2x_2 + \dots \\
 &= \sum_{\alpha} c_j(\alpha)x^{\alpha},
 \end{aligned}$$

where $x^{\alpha} = x_1^{\alpha_1} x_2^{\alpha_2} \dots x_n^{\alpha_n}$.

- Let E and F be normed spaces, $M \subseteq E$ and $N \subseteq F$ be nonempty open subsets, and $p : M \rightarrow N$ be a polynomial function.
 - If $T_{\bullet}$ and $S_{\bullet}$ are both sequences of bounded multilinear transformations associated to p , then for any nonnegative integer n , $T_n = 0$ iff $S_n = 0$.
 - The degree of p is a well-defined nonnegative integer.
 - There is a unique sequence of bounded multilinear transformations associated to p .
 - There is a unique sequence of bounded symmetric multilinear transformations associated to p .

- (e) For any nonnegative integer k , the k -th derivative $p^{(k)}$ of p is a polynomial function.
 - (f) $p \in C^\infty(M; N)$.
4. Let E and F be normed spaces, $M \subseteq E$ and $N \subseteq F$ be nonempty open subsets, $f : M \rightarrow N$ be a function and d be a nonnegative integer.
- (a) If f is a polynomial function of degree d , then for any integer $k \geq d + 1$ and for any $x \in M$, $f^{(k)}(x) = 0$.
 - (b) If for any $x \in M$, $f^{(d+1)}(x) = 0$, then f is a polynomial function of degree d .
 - (c) If M is path connected and for any $x \in M$, $f^{(d+1)}(x) = 0$, then f is a polynomial function of degree d .

3 Generative AI and Computer Algebra Systems Regulations

This section applies if you decide to use either a **generative AI tool** or a **computer algebra system** for this problem set. If not, you may skip this section.

3.1 Submission Requirements: Interaction Logs

If you use such tools, you must submit a complete log of your interactions. This is a mandatory part of your submission.

You can fulfill this requirement by either:

- **Listing the URLs** for your interactions in the appropriate field of the submission form, or

- **Attaching PDFs** containing the full interaction history to your Gradescope submission.

How to Obtain Logs You can generate the necessary logs using one of the following methods:

- **Shared Links:** Some AI tools (e.g., [ChatGPT](#)) can generate a unique, shareable URL for a conversation.
- **Web Archives:** Use a service like the [Wayback Machine](#) (see their "[Save Page Now](#)" documentation) to create a permanent, timestamped archive of your chat. There are also third-party browser extensions that are designed to extract chat logs from generative tools.
- **Screenshots:** Manually take screenshots of the entire interaction and compile them into a single PDF file. You can use your operating system's built-in tools or a third-party application.

3.2 Chat Guidelines and Prompt Engineering

When using chatbots, you must adhere to the following guidelines:

- **Permitted:** You may copy and paste text from this problem set, the textbook, or other course materials into the chat to provide context.
- **Prohibited:** You may not directly ask for a complete solution to a problem.
- **Required:** You must begin your conversation with a "guardrails" prompt designed to structure the AI's responses. This practice is known as [prompt engineering](#), and it is your responsibility you

design an appropriate guardrails prompt. The staff will evaluate the reasonableness of your attempts based on the logs you submit.

Here are some guides regarding prompt engineering:

- <https://platform.openai.com/docs/guides/prompt-engineering>
- <https://developers.google.com/machine-learning/resources/prompt-eng>
- <https://www.ibm.com/topics/prompt-engineering>
- <https://aws.amazon.com/what-is/prompt-engineering/>

When in doubt, check with the course staff to ensure your approach is appropriate.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

<https://forms.gle/7VUFcFPhWKuhxmsW8>.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

<https://www.gradescope.com/courses/1073533/assignments/6451792>,

see the Gradescope [documentation](#) for instructions.

You will receive a zero for this assignment if you submit your work on Gradescope under the wrong assignment, even if you submit your work on time.

5 When to Submit

This problem set is due on October 24, 2025 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.